

Supplemental Material for: Non-Additive Strain–Doping Coupling in Quasi-Hexagonal C₆₀ Graphullerene

Xingqiang Chen and Qixing Wang
Department of Physics, Xiamen University, Xiamen 361005, China

I. PERIODIC SYNERGY ROWS ($n \geq 6$)

II. MARCUS REORGANIZATION ENERGIES

Polaron transport diagnostics are orthogonal to the total-energy synergy audit $\mathcal{S}$ in the main text.

III. HOPPING INTEGRAL J

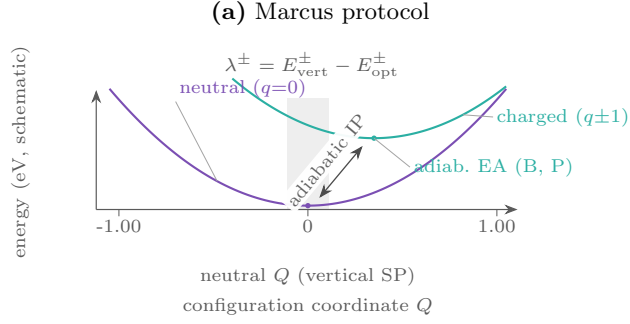
On the tetramer dimer grid, B substitution at $\epsilon=0$ delocalizes the charged-state orbital (IPR $75.0 \rightarrow 24.6$) while raising J from 26.6 to 46.4 meV; coupled B @ +3% gives $J = 37.2$ meV and IPR 23.1. The criterion $J > \lambda/2$ for polaron hopping dominance [1] is *not* met at PBE+D3 ($J < \lambda/2$), so a polaron transition is not confirmed on this grid.

[1] A. Capobianco, A. Breglia, P. Gentile, G. Capano, A. Asinari, G. Drera, L. Sangaletti, M. Cossi, and S. Casassa,

electron localization and mobility in monolayer fullerene networks, *Nano Lett.* **24**, 7891 (2024).

TABLE I. Supplemental periodic synergy rows at +3% ($n \geq 6$; 350 Ry production cutoff; completeness only).

n	Dopant	ϵ (%)	$\mathcal{S}$ (meV/atom)
6	B	+3	-7.76
6	N	+3	+2.38
6	P	+3	-31.89
8	B	+3	-11.26
8	N	+3	-2.30
8	P	+3	-29.94



(b) Adiabatic IP/EA and $\lambda^\pm$

Dopant	IP (eV)	EA (eV)	λ^+ (eV)	λ^- (eV)
Pristine	4.73	2.61	0.041	0.042
N	3.89	2.77	0.058	0.065
B	4.11	3.12	0.051	— [†]
P	4.07	2.96	0.200	0.315

[†]B channel: $\lambda^- < 0$ under fixed neutral-geometry vertical SP; not reported.

FIG. 1. Marcus reorganization energies from charged-state geometry optimizations and vertical single-points at the neutral geometry. Panel (a) schematically defines the vertical-adiabatic decomposition following Ref. 1; panel (b) tabulates adiabatic IP/EA and $\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$ for pristine and B/N/P tetramers. The electron-channel λ^- for B is unphysical under the fixed neutral-geometry vertical protocol and is excluded (Limitations).

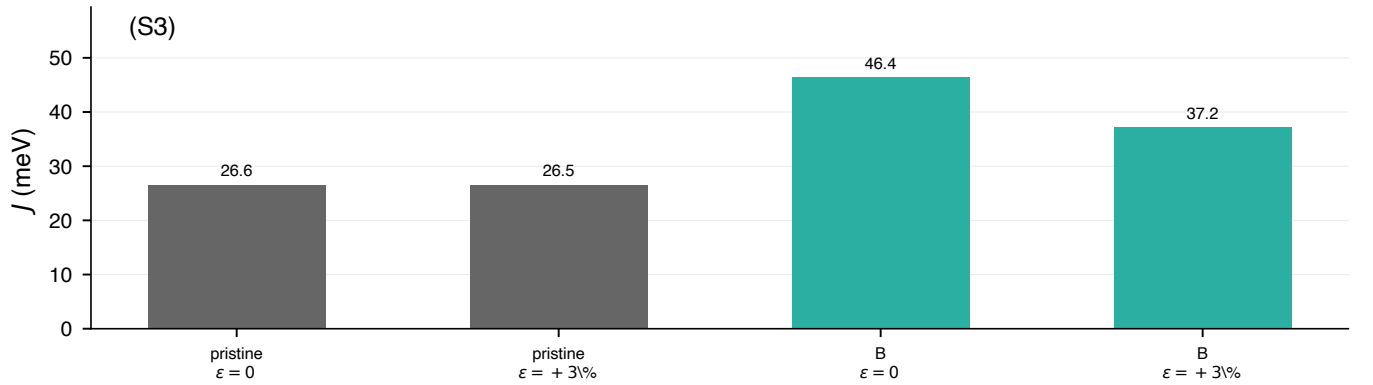


FIG. 2. Nearest-neighbor electronic coupling J on the tetramer dimer grid (pristine and B-doped cells at $\epsilon=0$ and +3%; PBE+D3). Companion inverse-participation-ratio values for the same four corners are tabulated in the open data release (Methods).